

# When GPUs Do Not Perform Best

## Evaluating UAD-2 OCTO DSPs for Low-Latency Streaming Workloads

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### Abstract

Modern graphics processors provide exceptional throughput when a workload exposes enough parallel work to amortize launch, synchronization, and data movement. Real-time audio and other streaming workloads often occupy the opposite regime: small buffers, persistent state, serial dependencies, and hard deadlines. This paper asks where an eight-DSP UAD-2 OCTO accelerator can outperform a commodity discrete GPU when performance is defined by end-to-end latency, tail latency, deadline misses, throughput, and energy, rather than peak floating-point rate alone. We specify matched workloads, correctness requirements, optimized baselines, measurement procedures, and a decision rule for identifying crossover regions. The present UAD stack initializes all command and response rings, receives a valid Linux-controlled framework response, and submits one integrity-protected resource independently to all eight DSPs. It also completes the captured 13-resource RealVerb pass on DSP0. Static analysis recovers the host runtime address-patch and resource-readback builders, although a bounded standalone readback produced no response. The inner executable, DSP-side relocation, entry-point, and buffer rules remain unresolved, so it has not completed a general-purpose program. Therefore, this version is a study design and does not claim a measured UAD performance advantage. Its central contribution is a falsifiable protocol that can support such a claim if, and only if, the resulting data satisfy the preregistered criteria.

**Evidence status.** The repository contains no completed UAD program and no completed UAD-versus-GPU comparison. All comparative outcomes in this manuscript are hypotheses. A results paper must replace this status with correctness-checked measurements.

## 1 Introduction

GPUs are designed to sustain high throughput across many concurrent threads. That design is compelling for large matrices, long convolutions, and highly parallel physical models. It does not imply that a GPU minimizes the completion time of every job. NVIDIA’s own optimization guidance emphasizes that host-device transfers must be included in application timing and that many small transfers should be batched [4]. Those costs are especially relevant when a real-time system must process a small buffer and return it before the next buffer arrives.

Prior audio-GPU studies already suggest a crossover rather than a universal winner. Renney, Mitchell, and Gaster reported a hardware-dependent baseline GPU overhead near 0.1 ms and found that feasible buffer sizes depended on sample rate, latency, and jitter constraints [6]. A later comparison of CPU, GPU, and FPGA implementations of modal reverberation showed both the very high throughput available from a GPU and meaningful variation in per-buffer latency [3]. These findings motivate a workload-centered study.

The target platform is a Universal Audio UAD-2 OCTO PCIe card. Universal Audio describes the OCTO as an eight-SHARC accelerator [9]. Static and high-resolution optical evidence confirms all eight processors as ADSP-21469 KBCZ-00 devices, for which Analog Devices specifies up to a 450 MHz floating-point core and 2700 MFLOP/s peak performance per processor [2]. Peak rates alone are not the proposed advantage. The hypothesis is that persistent DSP state, audio-oriented execution, and independent engines may benefit small, deadline-sensitive streams.

This study is designed to make that hypothesis difficult to confirm by accident. It compares equivalent numerical work, includes optimized GPU paths, reports tail behavior rather than averages alone, and includes workloads expected to favor the GPU. The intended contribution is a map of the crossover boundary, not a collection of hand-picked wins.

**Contributions after execution is available.** The completed paper is intended to provide:

1. a reproducible UAD-2 OCTO execution path with bounded DMA, correctness checking, and per-DSP recovery;
2. matched UAD, GPU, and reference CPU implementations of small streaming kernels and high-parallelism controls;
3. end-to-end latency distributions, deadline-miss rates, throughput, energy, and one-to-eight-DSP scaling; and
4. an explicit crossover map over buffer size, channel count, workload statefulness, and GPU contention.

At present, the first contribution’s transport, valid-response, authenticated resource-loader targeting, and empty-engine recovery components are established. Entry-point execution and bounded program output are not [8].

## 2 Research Questions and Claims

**RQ1.** For which combinations of buffer size, channel count, and kernel does the UAD have lower end-to-end p99 latency than an optimized discrete GPU?

**RQ2.** At a fixed audio deadline, which platform has the lower miss rate and smaller worst-case scheduling

jitter?

**RQ3.** Where does the UAD provide lower energy per correct output buffer, and does that conclusion survive idle-power subtraction and uncertainty analysis?

**RQ4.** How does throughput scale from one to eight independent UAD DSPs, and what shared transport bottlenecks limit that scaling?

**RQ5.** At what point do increasing arithmetic intensity and parallelism make the GPU the unambiguous winner?

The primary hypothesis is narrow: the UAD may win for small, stateful, deadline-sensitive streams whose work cannot fully occupy a GPU. The reciprocal hypothesis is equally important: a GPU should win as buffers, independent channels, filter length, or model size increase. A credible paper must report both regions.

### 3 Tested Platform and Current Evidence

The tested endpoint reports PCI identity `1a00:0002`, subsystem `1a00:0005`, a 64 KiB BAR, a PCIe 2.5 GT/s x1 link, and eight DSP-ready locations. The reverse-engineered host path publishes 16 command and response rings using 64 IOMMU-contained pages. It can start all eight engines and reset each empty engine independently. These observations are tied to one Rev. 5 card and should not be generalized to older UAD-2 endpoints.

Table 1 separates transport evidence from performance evidence. The distinction is essential: a moving ring index proves dequeue behavior, not correct program execution or low latency.

An isolated Windows reference run used the exact UAD 11.0.1 media, Secure Boot, TPM, and the signed OCTO driver. The official updater’s extended header, 624 full-page payload descriptors, 2,192-byte tail descriptor, and response descriptor were all consumed. After RTC-backed cold recovery, the driver again published all 16 rings and enabled all eight DSP engines. Pausing at the exact response-read-index boundaries showed zero-filled target pages, so descriptor consumption is not counted as a valid runtime response or program result [8].

A subsequent official plug-in-host experiment crossed that boundary. Four responses used header `0x80030302`, carried request-correlated tokens, and shared one 768-dword body. Static analysis maps its wire values to Demo not started, Authorized, Authorized, and Auth update required. The two authorized encodings collapse to the same host-visible state. RealVerb-Pro then reached an all-zero response for its first captured object, resource `0x12b`.

Linux later adopted the resident framework with fresh IOMMU mappings. Query `0x00260001` returned header `0x800c0005` in 1 ms. The exact 460-byte `0x12b` object returned `80070004 00000000 0000012b 00010073`, also in 1 ms. Single-bit changes at both ends of its 48-byte preserved prefix and at four locations in its 384-byte

trailing core all produced structured rejection statuses. Finally, the unchanged object succeeded independently on DSP0 through DSP7 with target-specific interrupt masks and no non-target ring-index movement. These results prove response transport, whole-body integrity enforcement, and loader targeting. They do not reveal the clear executable or activate an entry point [8].

Static analysis of the plug-in runtime further identifies a fixed `0x0af0`-byte allocation record, 16-byte memory-spec entries, 8-byte readback specs, command class `0x00150000` for mapped-address patching, and command `0x000c0004` for resource readback. A Linux trial issued the fixed four-dword readback at the accepted `0x12b` mapping. The command was consumed, but its six-dword response descriptor remained untouched through the deadline. Thus the host ABI is known, while a complete **Process** transaction or activation state remains a prerequisite candidate [8].

A subsequent bounded replay submitted the complete first RealVerb resource pass. All 13 objects returned exact resource- and command-correlated `0x80070004` completions in 1 to 5 ms, including three objects carried by two page-bounded DMA descriptors. A fixed readback after the thirteenth success was again consumed without a response. Incomplete resource loading is therefore ruled out, but no process entry or output is established [8].

The complete-pass state was not reusable. A later resource reload and query `0x00260001` were consumed without responses. All 13 exact pool-zero unload commands were then consumed, and the isolated per-DSP reset sequence passed, but the runtime service remained unanswered. No private-resource zeroing or memory-spec update was submitted. Fresh official plug-in activation is therefore a prerequisite for the next allocation trial [8].

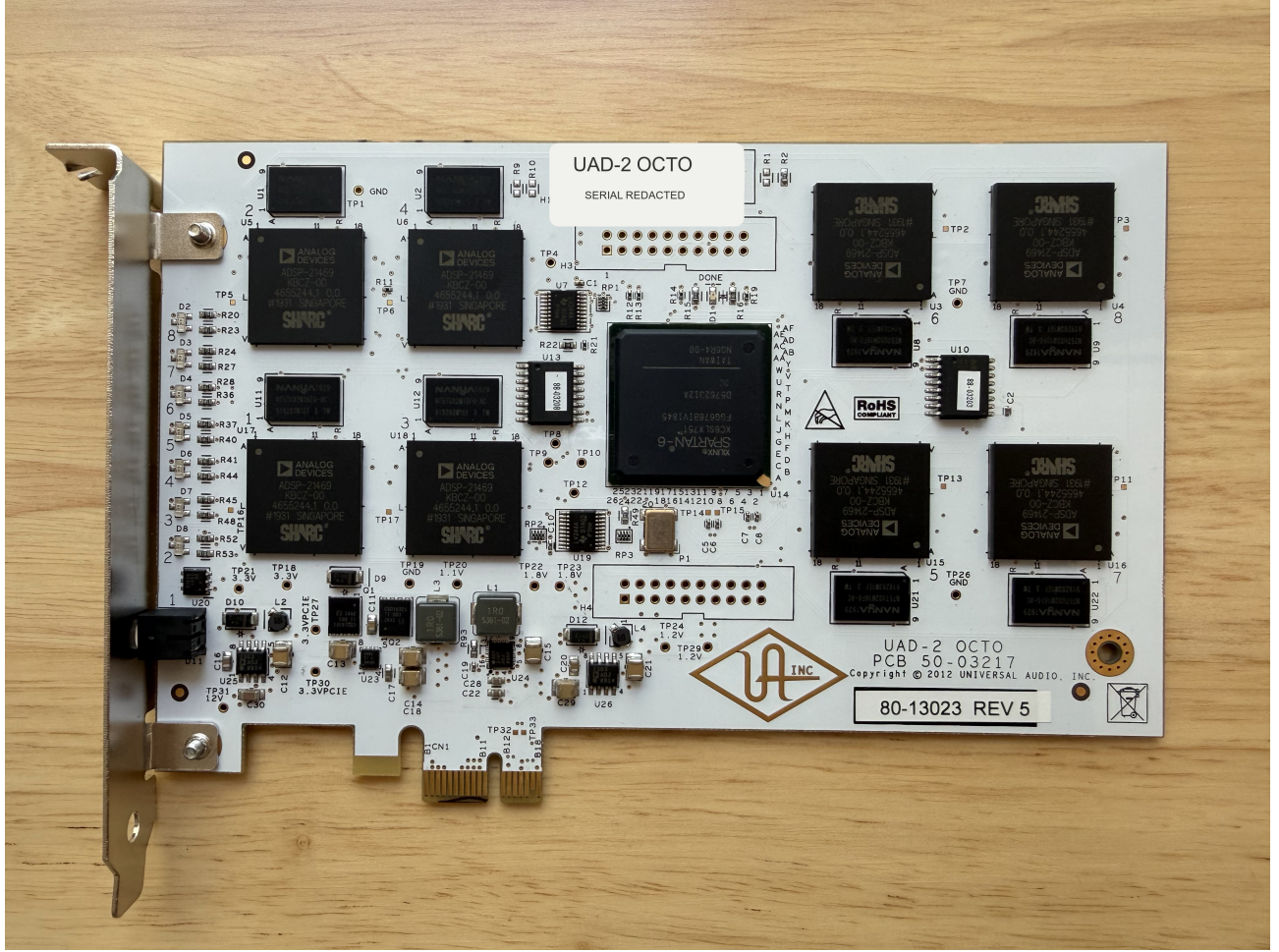
The processor identification is now direct at the component level. All eight packages visibly carry **ADSP-21469** and **KBCZ-00**; the central FPGA visibly carries **XC6SLX75T** and **FGG676**. The Analog Devices data sheet identifies the processor package as a 324-ball **CSP\_BGA** and specifies up to 450 MHz, while AMD lists the observed FPGA/package combination [2, 1]. The DSP speed ordering suffix is not visible, so the exact board clock remains unmeasured.

### 4 Comparison Model

The primary comparison is host-visible, end-to-end completion, as shown in Figure 2. Both platforms receive the same input samples and produce the same output layout. Timing includes every operation required for the host to submit one buffer and safely consume its result. Initialization, program loading, memory allocation, and one-time compilation are reported separately rather than charged to every buffer.

Two secondary modes explain where time is spent:

|                              |                                                                              |
|------------------------------|------------------------------------------------------------------------------|
| <b>Resident-stream mode.</b> | Inputs and outputs remain in device-accessible buffers across a long stream. |
|------------------------------|------------------------------------------------------------------------------|



**Figure 1:** The UAD-2 OCTO Rev. 5 board used by the project. The photograph optically resolves all eight ADSP-21469 KBCZ-00 SHARC packages and the central XC6SLX75T FGG676 Spartan-6 FPGA. The serial-label area is redacted in the public artifact.

Only the synchronization required by a real producer and consumer is retained. Device timestamps isolate kernel execution. This mode diagnoses compute efficiency but cannot support an end-to-end latency claim.

#### Device-only mode.

The GPU baseline must use pinned host memory, asynchronous copies where applicable, a persistent stream, and a CUDA Graph or equivalent low-overhead submission path. A simple launch-and-copy implementation may be reported as a secondary baseline, but it cannot be the primary GPU result. The reference CPU implementation is included to reveal cases in which neither accelerator is a sensible offload target.

## 5 Workload Suite

The suite in Table 2 spans the expected UAD-favorable regime and explicit GPU-favorable controls. Each kernel uses single-precision input and output unless a platform requires a documented alternative. Coefficients and initial

state are identical.

The primary audio operating points are 48 kHz and 96 kHz with buffer sizes 16, 32, 64, 128, and 256 samples. The full sweep extends from 1 to 1024 samples to expose the crossover shape. Independent channels are assigned consistently: first within one DSP, then across one through eight DSPs. The assignment policy must be fixed before timing begins.

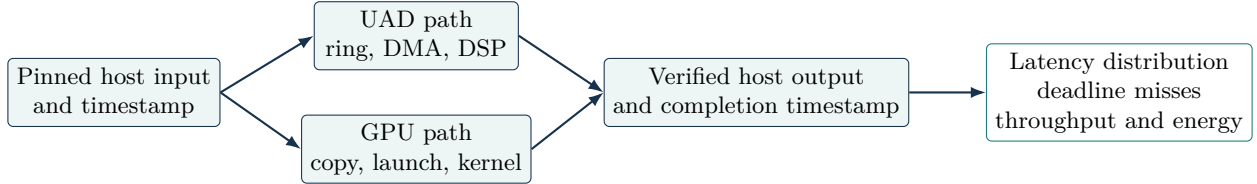
The serial channel chain is particularly important. A GPU can be penalized artificially by launching every stage separately. The primary GPU implementation must fuse compatible stages or capture the complete chain in a graph. The UAD implementation must likewise execute the chain as one persistent program when possible. The comparison is between best practical pipelines, not between deliberately unequal APIs.

## 6 Correctness and Equivalence

Performance is recorded only after correctness passes. A double-precision CPU implementation generates the

**Table 1:** Evidence available before the proposed benchmark campaign.

| Capability                                  | Status       | Interpretation                                                                                                                                                  |
|---------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exact PCI endpoint and isolated IOMMU group | Confirmed    | The tested card can be identified and its DMA can be contained.                                                                                                 |
| All-eight ring and DMA transport startup    | Confirmed    | All 16 rings and 64 pages can be published and recovered.                                                                                                       |
| Per-DSP reset on an empty transport         | Confirmed    | Each engine can be pulsed and returned to its baseline without changing the other seven.                                                                        |
| Valid runtime response                      | Confirmed    | Linux query 026 returns the exact five-dword response form in 1 ms; the official path also returns a decoded 768-entry authorization table.                     |
| Authenticated resource transport            | Confirmed    | Exact resource 0x12b succeeds on DSP0 through DSP7; all 13 resources in the captured RealVerb pass succeed on DSP0; sampled one-bit changes are rejected.       |
| Host runtime address patch and readback ABI | Confirmed    | The official host resolves selected-resource offsets through 0x00150000 and constructs 0x000c0004 readbacks; a standalone bounded request produced no response. |
| General-purpose program and bounded output  | Not achieved | No UAD workload can yet be compared for correctness or speed.                                                                                                   |
| GPU comparison dataset                      | Not achieved | A preliminary affine baseline does not provide a matched UAD result, power trace, or crossover dataset.                                                         |

**Figure 2:** Primary end-to-end comparison. Timing begins before platform submission and ends only after the output is visible to the host and verified. Device-only timing is reported separately and cannot replace this path.

reference output. Elementwise kernels pass when

$$\max_i |y_i - \hat{y}_i| \leq 10^{-5} \max(1, |\hat{y}_i|), \quad (1)$$

where  $\hat{y}$  is the reference. Stateful and FFT-based kernels additionally report normalized root-mean-square error and must satisfy a tolerance fixed before the benchmark. Denormal handling, rounding mode, fused operations, and fast-math flags are recorded. If a platform uses an approximation that changes quality, its performance is not placed in the same comparison cell.

Input sets include silence, impulses, steps, full-scale alternating values, logarithmic sweeps, deterministic pseudo-random samples, NaNs, infinities, and subnormals where the kernel contract permits them. State is reset between independent trials and preserved between adjacent buffers within a stream.

## 7 Metrics and Instrumentation

For buffer size  $B$  at sample rate  $f_s$ , the real-time deadline is

$$D = \frac{B}{f_s}. \quad (2)$$

Let  $L_i$  be the host-visible completion latency of buffer  $i$ . Deadline slack is  $D - L_i$ , and the empirical miss rate is

$$M = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[L_i > D]. \quad (3)$$

Every configuration reports median, p95, p99, p99.9, maximum, median absolute deviation, throughput, host CPU utilization, and miss rate. Means may be shown but are never the only latency statistic. Raw monotonic timestamps are retained in compressed machine-readable files.

The primary latency speedup is

$$S_{99} = \frac{L_{99, \text{GPU}}}{L_{99, \text{UAD}}}. \quad (4)$$

Thus,  $S_{99} > 1$  favors the UAD. Confidence intervals are computed by a session-aware bootstrap so that adjacent buffers are not treated as fully independent. Results are stratified by idle GPU, GPU with a display workload, and GPU with a controlled background compute workload. The idle, dedicated GPU case remains the primary baseline.

Device energy is measured with external PCIe power instrumentation when available. Whole-system energy is

**Table 2:** Preregistered workloads. “Purpose” describes the architectural question, not a predicted winner.

| Workload                     | Parameter sweep                                              | Work per output sample                    | Purpose                                                                  |
|------------------------------|--------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------|
| Identity round trip          | Buffers 1 to 1024; 1, 2, 8, and 32 channels                  | Copy input to output                      | Measures the platform and synchronization floor.                         |
| Affine transform             | Same buffers and channels                                    | $y = ax + b$                              | First bounded program; low arithmetic intensity.                         |
| Biquad cascade               | 1, 4, 8, and 16 sections per channel                         | Stateful recursive filter                 | Tests serial dependence and persistent local state.                      |
| Dynamics processor           | Envelope follower plus gain curve; attack and release sweeps | Stateful control and nonlinear arithmetic | Represents compressors and limiters without vendor code.                 |
| Short FIR                    | 8, 32, 128, and 512 taps                                     | Sliding dot product                       | Locates the transition from short stateful work to parallel convolution. |
| Serial channel chain         | Gain, four biquads, dynamics, delay, and mix                 | Multiple dependent stages                 | Tests whether one persistent program avoids repeated dispatch overhead.  |
| Long partitioned convolution | Impulses from 4096 to 131072 taps                            | Batched FFT and multiply-accumulate       | GPU-favorable audio control and crossover anchor.                        |
| Dense matrix multiply        | Square FP32 matrices from 64 to 2048                         | Highly parallel $O(n^3)$ work             | Out-of-domain throughput control expected to favor the GPU.              |

measured at the wall as a secondary metric. Idle power is recorded in the same thermal state and subtracted only in a separately labeled dynamic-energy result. Driver-reported instantaneous GPU power is not treated as equivalent to a high-rate external measurement.

## 8 Experimental Procedure

Each session records the CPU, GPU, UAD identity, PCIe topology, firmware and driver versions, kernel, compiler flags, clock policy, thermal state, and background services. The UAD and GPU must be installed in the same host for the primary comparison unless slot bandwidth or topology makes that arrangement unfair; any separate-host result is labeled as such.

For each configuration:

1. verify the platform baseline and allocate all buffers;
2. load or capture the program, then execute 2,000 un-timed warm-up buffers;
3. collect 20,000 buffers for the full parameter sweep;
4. collect at least 100,000 buffers in each of five independent sessions for the preregistered primary operating points;
5. verify every output, capture faults and deadline misses, then restore the platform baseline; and
6. rerun the primary points in an interleaved platform order to reduce time-of-day and thermal bias.

The primary points are the affine transform, four-section biquad, dynamics processor, and serial channel chain at 48 kHz; buffers of 32 and 64 samples; and 1 and 8 channels. These points are fixed before any comparison data are examined. The broader sweep is used to estimate crossover surfaces, not to retroactively select the headline

result.

## 9 Decision Rule

A configuration is labeled a UAD latency win only when all of the following hold:

1. both outputs meet the same preregistered correctness threshold;
2. the lower bound of the 95 percent confidence interval for  $S_{99}$  is greater than one;
3. the UAD deadline-miss rate is no higher than the GPU rate within the stated uncertainty; and
4. the result reproduces across all five sessions without a change in code, clocks, or exclusion rules.

Energy, throughput, and isolation wins are separate claims with separate denominators. A lower p99 latency does not imply greater throughput, and a reset-isolation advantage is not described as faster computation.

## 10 Results Status and Required Presentation

No comparative result exists yet. The current API intentionally returns `-EOPNOTSUPP` for buffer allocation, program loading, submission, and waiting. A valid response and authenticated resource completion are no longer the blockers, and the complete captured resource pass now succeeds. The remaining gate is a decoded OCTO executable with proven segments, DSP-side relocations, entry point, reserved ranges, and program-visible buffer semantics. The recovered host address-patch layer does not satisfy this gate. It must first run as a bounded DSP0

heartbeat, followed by the affine transform. The results section must remain empty of performance claims until those gates pass.

The current live card also requires official runtime re-activation before the allocation experiment can resume. Transport readiness and reset success are not treated as application-service readiness.

When measurements are available, the paper should present four views:

1. a correctness table with numerical error for every implementation;
2. latency complementary cumulative distribution plots at primary points;
3. heat maps of  $S_{99}$  over buffer size and channel count for each workload; and
4. scaling and energy plots with uncertainty intervals.

The headline table must include UAD losses and ties as well as wins. Failed or unsupported UAD configurations remain in the denominator and are not silently removed.

## 11 Isolation and Robustness

The OCTO exposes eight engines with independently exercised empty-transport reset paths. Authenticated loader targeting is now proven on all eight engines: the selected engine consumes the exact object while all seven non-target ring index sets remain unchanged. Program-level execution isolation is not yet proven. After correctness and single-DSP performance are established, each DSP is tested under normal completion, program timeout, response timeout, process exit, malformed program rejection, and an IOMMU-contained fault. Status, ring indices, and output buffers for all other DSPs are recorded before and after each case.

Robustness is reported as availability and recovery behavior, not folded into kernel latency. A separate service-level experiment may compare the useful throughput retained when one UAD engine fails with the useful throughput of a GPU process under an analogous isolated failure. The failure models must be defined carefully because the architectures expose different reset scopes.

## 12 Threats to Validity

**One board and one revision.** All current hardware evidence comes from one OCTO Rev. 5 card. Replication on additional boards is required before making population-level reliability or performance claims.

**Processor identification.** The model and KBCZ-00 package marking are optically confirmed on all eight tested packages. The speed ordering suffix is not visible, and no core clock has been measured. Results must therefore separate component identity from catalog maximum-frequency specifications.

**Baseline selection.** “Regular GPU” is not a reproducible category. The paper must name the exact GPU, memory, PCIe link, driver, runtime, power limit, clocks, and whether it drives a display. A second GPU from another performance tier would improve external validity.

**Optimization asymmetry.** A mature vendor GPU stack and a new reverse-engineered DSP stack can receive unequal tuning effort. Source code, profiling evidence, and optimization logs should be published for both. Negative results caused by an incomplete UAD runtime cannot be interpreted as an architectural limit.

**Workload selection.** Small audio kernels are intentionally chosen to test the hypothesis. Long convolution and dense matrix controls prevent that choice from being presented as universal superiority.

**Timing and power resolution.** Coarse polling counters, driver timeouts, datasheet peak rates, and a reset duration that excludes time inside an ioctl are not performance measurements. Only synchronized end-to-end timestamps and calibrated energy instrumentation support the corresponding claims.

## 13 Related Work

Renney et al. provide a benchmark-oriented study of practical GPU audio processing and demonstrate the importance of buffer choice [6]. Michon et al. compare CPU, GPU, and FPGA modal reverberation implementations, showing both GPU scale and per-buffer latency variation [3]. NVIDIA’s guidance supplies the optimization requirements for host-device data movement and workload sizing [4].

Open Apollo and the experimental `stepbrobd/uad2` driver provide useful transport hypotheses for related hardware [5, 7]. They do not establish general-purpose execution or performance on the exact OCTO endpoint. The accompanying project therefore keeps related-device constants separate until the tested card or a hash-locked official binary confirms them [8].

## 14 Conclusion

The scientifically useful question is not whether a UAD is faster than a GPU in general. It is where the crossover occurs when both systems execute the same correct work under the same real-time contract. The most plausible UAD region contains small, persistent, stateful streams evaluated by p99 latency and deadline misses. The most plausible GPU region contains large, parallel, compute-dense workloads. This manuscript defines how to locate that boundary without converting an architectural intuition into an unsupported result.

The paper is ready to accumulate comparative measurements only after a harmless program produces bounded output and the result survives reset and IOMMU cleanup. Valid Linux runtime and resource-success responses have now been observed, body-wide integrity enforcement is demonstrated, and authenticated loader targeting passes on all eight DSPs. All 13 resources in the captured RealVerb pass also succeed on DSP0. The host runtime address-patch and readback builders are also recovered, but bounded standalone readback did not produce a response, and the current runtime requires fresh official activation. Until the inner executable, DSP-side relocation, entry-point, and buffer rules are recovered, the defensible finding is the transport, resource-loader, and isolation foundation, not performance superiority.

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